

Chengxi Luo

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Legally authorized to work in the U.S. without restriction

EDUCATION

University at Buffalo, State University of New York Buffalo, NY
Bachelor of Science in Computer Science • Expected: May 2028 *August 2026 - Present*

Bernard M. Baruch College, City University of New York New York, NY
Coursework in Computer Science • Dean's List (All Semesters) • GPA: 3.91 / 4.00 *January 2025 - May 2026*

EXPERIENCE

Research Assistant | *University at Buffalo, DRONES lab* *September 2026 - Present*

- Integrated Fast LiDAR-Inertial Odometry (Fast-LIO) into the robot system to improve localization and mapping.
- Redesigned and implemented the robot's motion command handling, mitigating conflicts from various control inputs.
- Manage the project's core ROS packages on GitHub, triaging and resolving issues to keep the software stable.

Committer (Open-Source Maintainer) | *Apache Iggy (Apache Software Foundation)* *June 2025 - Present*

- Elected to Committer status by the project's Project Management Committee (PMC), merging 40+ pull requests and reviewing 30+ additional PRs across Go SDK, Rust server, and test infrastructure, while mentoring new contributors.
- Implemented core Go SDK features — leader redirection, context.Context support, and a unified Command interface with per-type Marshal methods — standardizing request serialization across all client operations.
- Redesigned the TCP connection lifecycle (establish, retry, timeout, teardown) and cluster advertised-address handling, improving reconnection reliability under network failures.
- Built the Go SDK's BDD test suite and CI/CD pipeline, closing a major gap in release-quality assurance.

PROJECTS

Serify | *Creator* *July 2026 - Present*

- Built an open-source cross-language serialization conformance framework that verifies byte-identical (or semantically identical) output across 9 programming-language implementations from a single shared schema.
- Designed a YAML-based schema DSL, NDJSON worker protocol, and Go CLI to drive polyglot worker processes through a common test suite and diff their results.
- Implemented detection of unsafe serialization behaviors — mutation, non-determinism, and buffer aliasing — catching correctness bugs that simple byte-comparison testing would miss.

Better CUNY Chrome Extension | *Creator* *May 2025 - January 2026*

- Built a crawler to extract CUNY student schedules and export them as CSV files for one-click Google Calendar import, growing to 511+ installs with a core base of 37+ active users.
- Automated the CUNY student login and schedule-scraping workflow via browser scripting, turning a multi-step manual export into a single automated run.

HOLD Cloud Desktop Service Platform | *Backend Developer & DevOps* *January 2024 - April 2024*

- Implemented container desktop operations including create, start, delete, scale, file transfer, and VNC access.
- Deployed and managed services and databases across VMs on Tencent Cloud and Alibaba Cloud infrastructure, building live server-status detection to automatically flag instance failures.
- Built a real-time distributed monitoring system tracking CPU, memory, and network metrics across all deployed VMs, using Redis locks to prevent race conditions and duplicate monitoring tasks.

SKILLS

Programming Languages: Go, Rust, TypeScript/JavaScript, Python, Java, C++, Kotlin
Backend: Gin, gRPC, SQL, Redis, etcd, Node.js, Spring Boot, Message Queue, TCP, HTTP, QUIC, WebSocket
Data: pandas, Selenium, Scrapy
Frontend: React, Next.js, Material UI, Android Jetpack Compose, HTML, CSS, Browser Extension Development
Technologies: Git, Docker, Linux, GitHub Actions, CI/CD, REST API, Robot Operating System (ROS)
Computer Science Fundamentals: Data Structures, Algorithms, Object-Oriented Programming
Practices: Test Driven Development (TDD), Behavior Driven Development (BDD), Concurrent Programming (Goroutines, Channels), Unit Testing
Cloud Platforms: Tencent Cloud, Alibaba Cloud (VM management, object storage, networking, cloud disk)